Differentiation: rates and tangents

Lecture and worked examples

Chapter path

Objectives. Interpret a rate; approximate a derivative at a point; determine a tangent; understand why differentiability fails.

Prerequisites. Function values, slope of a line, algebraic manipulation, square root and absolute value.

Yearly reference point. 3-12 November: 8 hours

Rate of change and secants

For $a \neq b$ in the domain of f, the rate of change between a and b is $\frac{f(b) - f(a)}{b - a}$. It is the slope of the secant through the corresponding points. In context, the unit of the rate is the unit of f divided by that of the variable.

Rate of change and secants

For $f(x) = x^2$, between 1 and 3, the rate is $\dfrac{9-1}{3-1} = 4$. Between 1 and $1+h$, it is $2+h$. The slopes for $h = 0.1$ and $h = -0.1$ are 2.1 and 1.9.

Rate of change and secants

The rate is not $\frac{f(b)}{f(a)}$; h represents a gap, positive or negative depending on the domain.

Rate of change and secants

Calculate the rate of $f(x) = x^2 + 3x$ between 2 and $2 + h$.

Derivative at a point

f is differentiable at a when the rate $\frac{[f(a+h)-f(a)]}{h}$ tends to a real number as h tends to 0, within the domain. This real number is $f'(a)$. An intuitive approach to the limit is used here. In kinematics, if f gives a position in metres and time is in seconds, the derivative at a point represents an instantaneous velocity in $\mathrm{m\,s^{-1}}$.

Derivative at a point

For $s(t) = t^2 + 3t$, the rate around 2 is $7 + h$, so $s'(2) = 7\ \mathrm{m\,s^{-1}}$. The average velocity on $[2, 3]$ is $8\ \mathrm{m\,s^{-1}}$; it is not the velocity at time 2.

Derivative at a point

We do not replace h by 0 in an unsimplified quotient: we would get a division by 0.

Derivative at a point

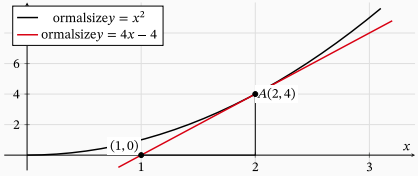
For $f(x) = 3x^2 - 2x$, calculate $f'(1)$ from the rate.

Tangent and affine approximation

If f is differentiable at a, the tangent at the point $(a, f(a))$ has equation $y = f(a) + f'(a)(x - a)$. For small h, we use $f(a + h) \approx f(a) + f'(a)h$. This local approximation does not by itself provide a bound on the error.

Tangent and affine approximation

For $f(x) = x^2$ at 2, tangent $y = 4 + 4(x-2) = 4x - 4$. We approximate 2.01^2 by $4 + 4(0.01) = 4.04$; the exact value is 4.0401, error $0.0001 = h^2$.



Tangent and affine approximation

A tangent may cross the curve again elsewhere. “Only one common point” is not a definition.

Tangent and affine approximation

Given $f(3) = 5$ and $f'(3) = -2$. Write the tangent and approximate $f(3.04)$.

When differentiability fails

The square root function is defined on $[0, +\infty)$ but is not differentiable at 0. The absolute value function is defined on $\mathbb{R}$ but is not differentiable at 0. Being defined, or having a continuous curve, therefore does not guarantee differentiability.

When differentiability fails

For $|x|$, the secants from the origin all have slope 1 on the right and -1 on the left. The corner point does not have a unique tangent with finite slope. For the square root, the limiting direction is vertical; there is no real derivative.

When differentiability fails

A very large slope is not a proof of non-differentiability; the behaviour of the rate must be explained.

When differentiability fails

Justify that $g(x) = |x - 2|$ is not differentiable at 2. Also study the differentiability of $x \mapsto \sqrt{x}$ at 0 through its rate.